

Lecture 19

ChE 467/596-012 Polymer Rheology

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of

Recap & Agenda

- Last class: Multiphase systems—foams & emulsions
- Today: Start on suspensions

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Multiphase Systems

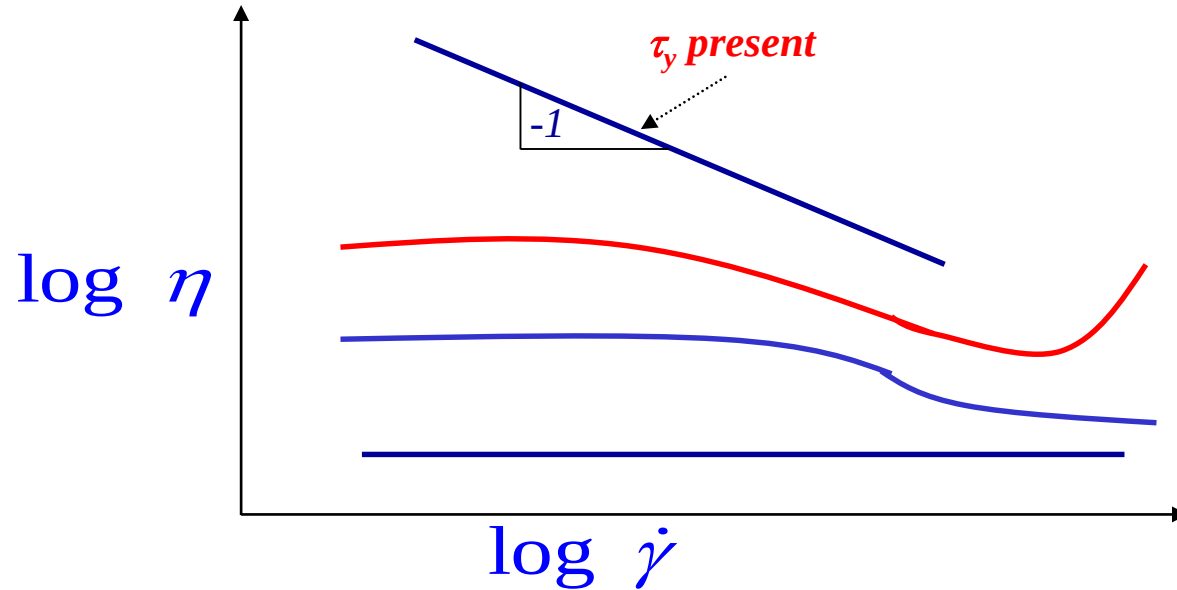
- *Air in Liquid* ————— *Foams*
- *Liquid in Liquid* ————— *Emulsion*
- *Solid in Liquid* ————— *Suspension & Filled systems*

Foams and emulsions exhibit similar characteristics

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Suspensions

Spheres (solid) can generally show a range of steady shear behavior



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Dilute Suspensions

Einstein (1906-1911)

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Assumptions:

1. **Creeping flow: no inertial effect*
2. *Surrounding fluid is Newtonian*
3. **No particle settling*
4. **No particle migration*
5. **No wall effect*
6. *Rigid spheres*
7. *No slip between particle and fluid*

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Dilute Suspensions, cont'd

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Assumption 1:

Creeping flow: no inertial effects

Re[#] has to be small (<<0.1)

If Re[#] is not small

$$\eta = \eta_s \left[1 + \frac{5}{2} \phi + 1.34 \text{Re}_p^{3/2} \right]$$

$$\text{Re}_p = \frac{\rho \dot{\gamma} a^2}{\eta_s}$$

Assumption 5:

No Wall effect

Gap has to be >> 10 particle diameter

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Dilute Suspensions, cont'd

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Assumption 3:

No particle settling

$$\rho_s \cong \rho_p;$$

If $\rho_s \neq \rho_p$

Need to determine how long particles take to settle

$$t_{0.1h} = \frac{0.45\eta_s h}{\left| (\rho_p - \rho_s) \right| a^2 g}$$

a = particle radius

h = gap or height (Couette)

ρ_s = solvent viscosity

$t_{0.1h} \Rightarrow$ time required to settle 10% of rheometer gap or Couette height

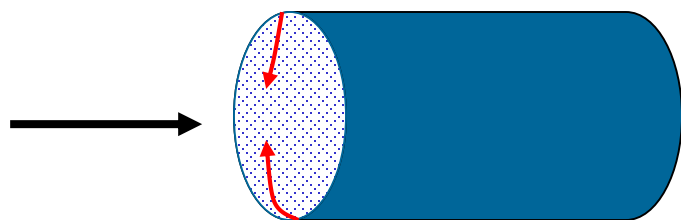
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Dilute Suspensions- cont'd

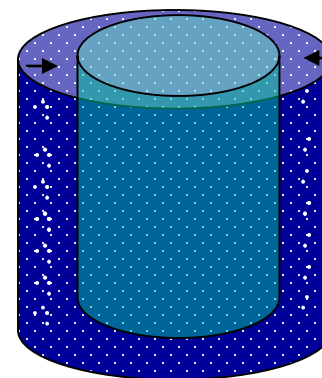
$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Assumption 4:

No particle migration



$$K = \frac{\rho_p \overline{v^2} a^4}{h k T}$$



$$\overline{v} = \frac{1}{2} v_{\text{max}} \text{ in Couette}$$

$$\overline{v} = \frac{2}{3} v_{\text{max}} \text{ in Poiseuille flow}$$

k Boltzman constant

T Temperature

If $K < 0.1$ Couette

$K < 0.01$ Poiseuille

Then we can neglect migration of particles

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Generalized Einstein's Equation

Batchelor & Schowalter

$$\underline{\underline{\tau}} = \underline{\underline{\tau}}_s + \underline{\underline{\tau}}_p$$

Stress contribution Stress contribution
from solvent from particles

$$\underline{\underline{\tau}}_p = \frac{1}{V} \int \text{around the particle}$$

$$\underline{\underline{\tau}} = 2\eta_s \left[1 + \frac{5}{2} \phi \right] \underline{\underline{D}}; \quad 2\underline{\underline{D}} \equiv \text{Rate of strain tensor}$$

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Einstein's Equation

As ϕ increase, particle start to interact with each other

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Generalized Einstein's Equation, cont'd

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$



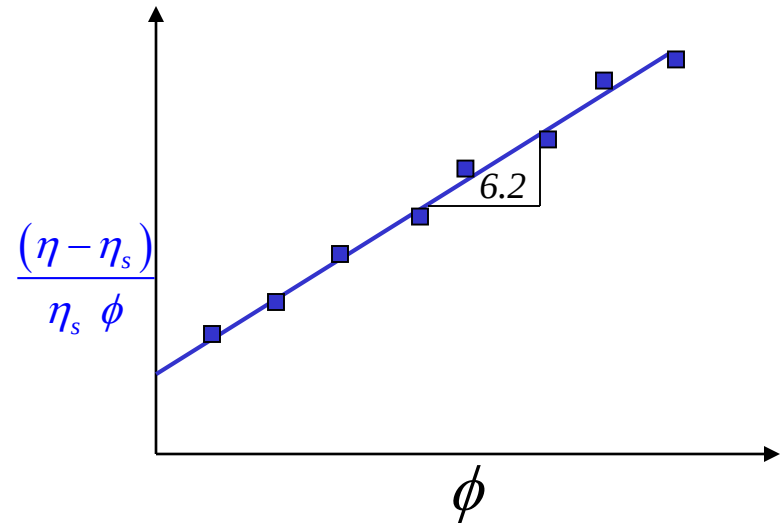
Takes into account interaction between particles

$$\eta = \eta_s \left[1 + \frac{5}{2} \phi + k\phi^2 + \dots \right]$$

Rearrange



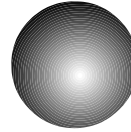
$$\frac{(\eta - \eta_s)}{\eta_s \phi} = \frac{5}{2} + k\phi$$



Experimentally $k=6.2$

Dilute Suspensions - cont'd

Non interacting spheres:



$$\eta = \eta_{\text{solvent viscosity}} \left[1 + \frac{5}{2} \phi \right] \quad \text{Einstein Equation}$$

(Note: In the original image, 'solvent viscosity' and 'Volume fraction of filler' are written in green below their respective terms in the equation.)

Deviation:

Spheres start interacting

$$\eta = \eta \left(1 + \frac{5}{2} \phi + 6.2 \phi^2 \right)$$

Non-submicron spheres => no colloidal forces present
As the sphere size $\leq 1 \mu\text{m}$, start having interaction between particles

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Definitions

Relative or reduced viscosity:

$$\eta_r = \frac{\eta}{\eta_s}$$

Specific Viscosity:

$$\eta_{\text{specific}} = \frac{\eta - \eta_s}{\eta_s}$$

Intrinsic Viscosity:

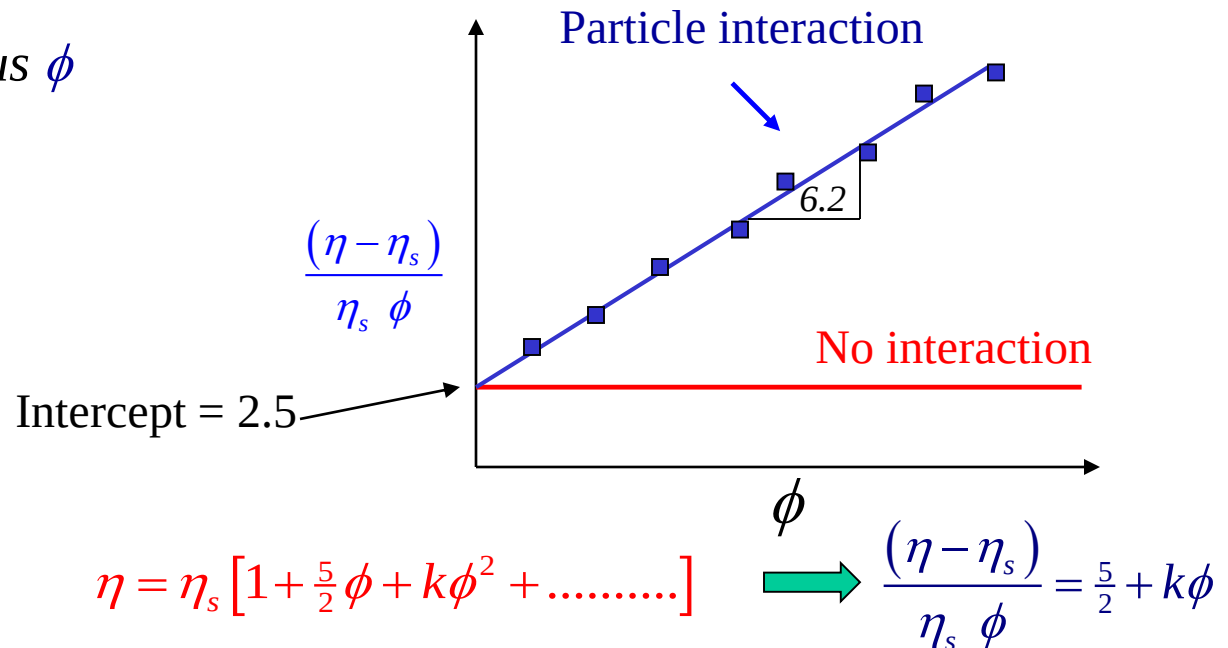
$$[\eta] = \lim_{\phi \rightarrow 0} \frac{\eta_{\text{specific}}}{\phi}$$

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In-class Problem

Problem: Given samples of different volume fraction of 2micron diameter glass beads in oil. Solvent & sample viscosities known for different ϕ .
How would you determine graphically if interactions between particles exist? What is the intrinsic viscosity of these suspensions?

Plot $\frac{(\eta - \eta_s)}{\eta_s \phi}$ versus ϕ



$$\frac{(\eta - \eta_s)}{\eta_s \phi} = \frac{5}{2} + k\phi$$

$[\eta]=2.5$ for spherical suspensions (From Einstein Equation)

In-class Problem

Problem: Given PS latex (1 micron diameter) in water ($\eta = 1$ cp); $\phi = 0.05$;
 $\rho_p = 1.05 \rho_s$.

- (a) Obtain viscosity of suspension
- (b) Determine if inertia, migration or settling is important in couette system. $L = 50$ mm, $R_i = 25$ mm; $R_o = 26$ mm; $\Omega < 100$ rad/s

$$\dot{\gamma} = \frac{\Omega R_i}{R_o - R_i} \text{ in couette}$$

$$\begin{aligned} (a) \quad \eta &= \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right] \\ &= 1.13 \text{ cp} \end{aligned}$$

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In-class Problem

Problem: Given PS latex in water ($\eta = 1$ cp); $\phi = 0.05$; $\rho_p = 1.05 \rho_s$.

- (a) Obtain viscosity of suspension
- (b) Determine if inertia, migration or settling is important in couette system. $L = 50$ mm, $R_i = 25$ mm; $R_o = 26$ mm; $\Omega < 100$ rad/s

$$\text{Re}_p = \frac{\rho_p \dot{\gamma} a^2}{\eta_s} = 6.6 \times 10^{-6} \Omega$$

$$\dot{\gamma} = \frac{\Omega R_i}{R_o - R_i}$$

So, inertia not a problem

$$t_{0.1h} = \frac{0.45 \eta_s h}{|(\rho_p - \rho_s)| a^2 g} = 1.8 \times 10^5 \text{ sec}$$

Settling not a problem

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In-class Problem

Problem: Given PS latex in water ($\eta = 1$ cp); $\phi = 0.05$; $\rho_p = 1.05 \rho_s$.

- (a) Obtain viscosity of suspension
- (b) Determine if inertia, migration or settling is important in couette system. $L = 50$ mm, $R_i = 25$ mm; $R_o = 26$ mm; $\Omega < 100$ rad/s

$$\dot{\gamma} = \frac{\Omega R_i}{R_o - R_i}$$

$$K = \frac{\rho_p \overline{v^2} a^4}{h k T} = 0.0025 \Omega^2$$

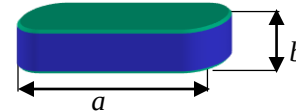
Migration negligible for $\Omega < 20$

$h =$ length of couette
avg. vel. $= \Omega R_i$

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Fibers - Dilute

Need to consider time scale of orientation/rotation of fiber



Aspect ratio, $r_p = \frac{a}{b}$

*Time scale of rotation / Brownian motion = $1/D_r$
(D_r is the rotary Brownian diffusion coefficient)*

$$D_r \cong \frac{kT}{8\pi\eta_s a^3}$$

For Spheres

$$D_r \cong \frac{3kT(\ln r_p - \frac{1}{2})}{8\pi\eta_s a^3}$$

For Prolate sphere

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Fibers-cont'd

Time scale of motion from flow $\sim \frac{1}{\dot{\gamma}}$

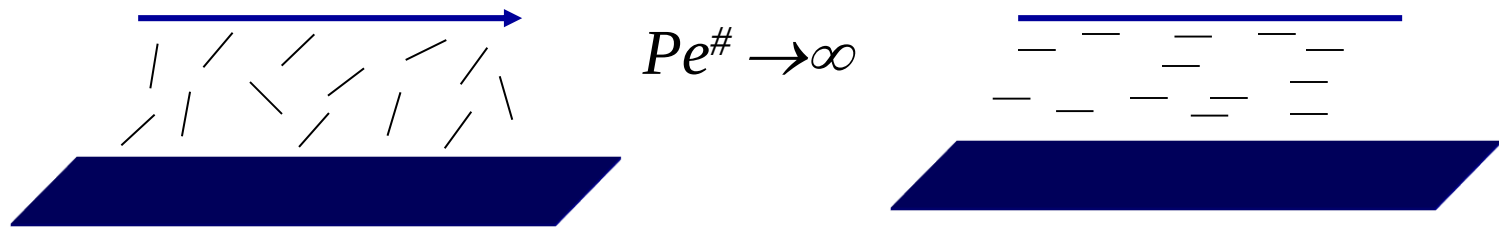
Define Peclet #

$$Pe^{\#} = \frac{\dot{\gamma}}{D_R}$$

$Pe^{\#} \rightarrow 0$ *Brownian motion dominates*

$Pe^{\#} \rightarrow \infty$ *Convective motion dominates*

As $Pe^{\#} \rightarrow \infty$, fibers orient along the flow direction

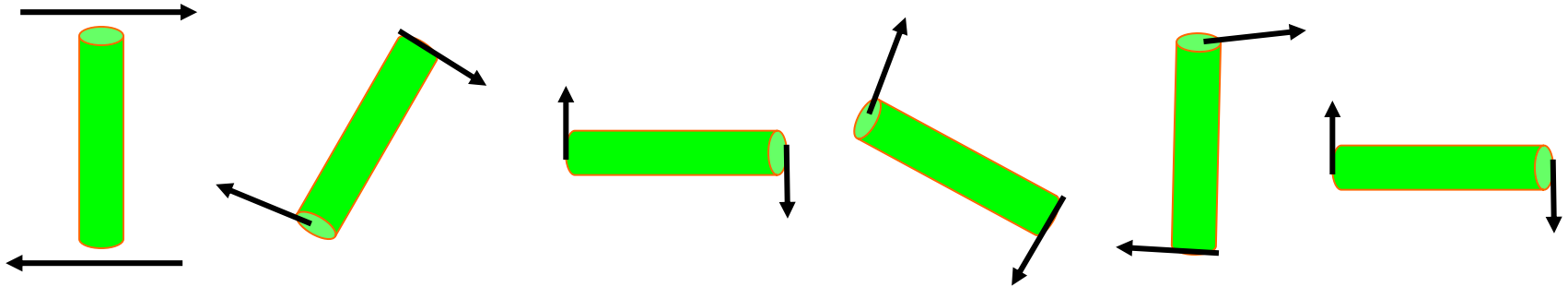


Reality: Fibers do orient but also rotate very slowly

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Fibers-cont'd

Fibers do orient but also rotate/tumble very slowly



Slow Rotation: Time period for 1 oscillation

$$T_{period} = \frac{2\pi}{\dot{\gamma}} \left(r_p + \frac{1}{r_p} \right)$$

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Fibers-cont'd

Orientation:

$$\tan \phi = r_p \tan \left(\frac{\dot{\gamma} t}{r_p + \frac{1}{r_p}} \right)$$

$\phi = 0$



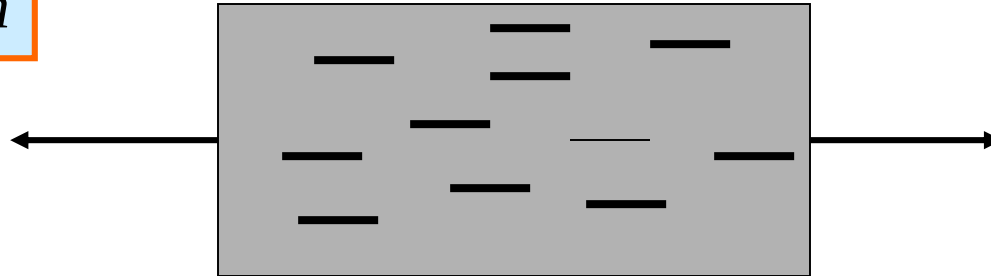
$\phi = 90$



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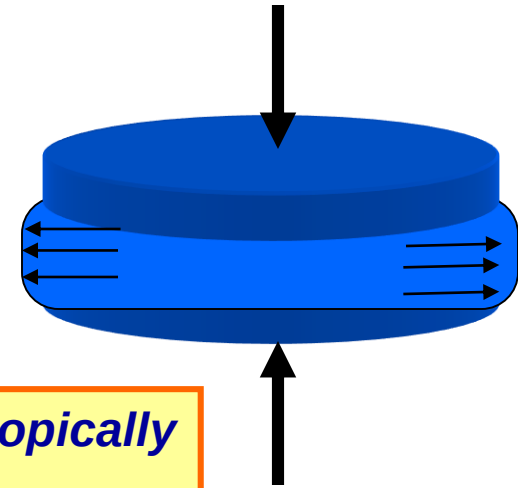
Fibers-cont'd

Extension



Ex: Squeezing flow between plates

Viscosity may be lower with fillers than without



- *you are applying extensional flow macroscopically*
- *have shear+extension at the micro level*

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Recap & Approach

A. Dilute Systems

1. spheres (Einstein eqn.,)

2. fibers (brief)

concept of rotation

concept of $Pe^{\#}$

B. Non-dilute Systems

1. Interparticle forces & potential energy

2. Interpret rheology

of

Non-dilute Suspensions

~1% can show complex non-dilute behavior

Relative motion of neighboring particles

→ hydrodynamic interaction forces

=> refers to interactions caused by flow

(present in all systems)

Lot of times, deal with small fillers ($< 1 \mu\text{m}$)

Small fillers => additional interactions

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Approach

B. Non-dilute Systems

1. Interparticle forces & potential energy

- **Attractive forces (dispersion)**
 - **van der Waals**
induced dipoles, possible in all systems
- **Repulsive forces**
 - **electrostatic**
 - **steric**

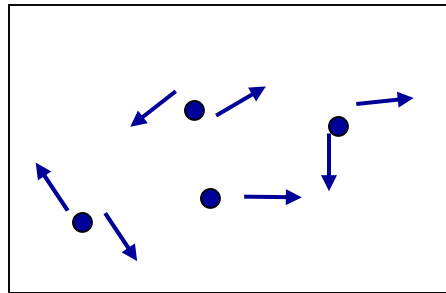
2. Interpret rheology (in terms of)

- **Hard spheres (Brownian)**
- **Stable system**
- **Flocculated system**

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Non-dilute Suspensions-Brownian Motion

- *Tends to randomize things*
- *Concerned with translational diffusion coefficient*



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Non-dilute Suspensions-Interparticle forces

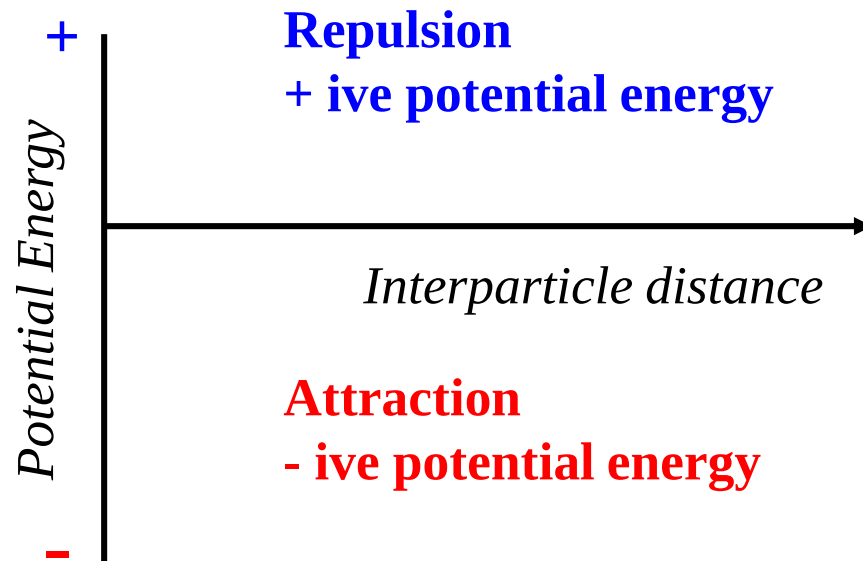
- *Several of these*
- *Magnitude dependent on distance between particles*
- *Attractive or repulsive*



As particles come close => repulsive forces in this case

Non-dilute Suspensions-Interparticle forces

Interparticle forces (attractive or repulsive) are often looked from the stand point of potential energy



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